

Lean port sheets

Rev F0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | D-044 lean ports A (PV) and B (battery) + insulation monitor, BUS- referenced

01 Supplies and commands

Live 24 V / 5 V / 3.3 V, BUS- tie, commands

File: sheets/01_lean_io.kicad_sch

02 Port A power path (pv)

Varistor Y + monitor

File: sheets/02_leanA_power.kicad_sch

03 Port A sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/03_leanA_ctrl.kicad_sch

04 Port A coil drives

Contactor: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/04_leanA_drive.kicad_sch

05 Port B power path (battery)

Varistor Y + monitor, aR fuses

File: sheets/05_leanB_power.kicad_sch

06 Port B sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/06_leanB_ctrl.kicad_sch

07 Port B coil drives

Contactor + precharge relay: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/07_leanB_drive.kicad_sch

08 Insulation monitor

PE divider, two switched 992 k strings through CA-IS3417WT

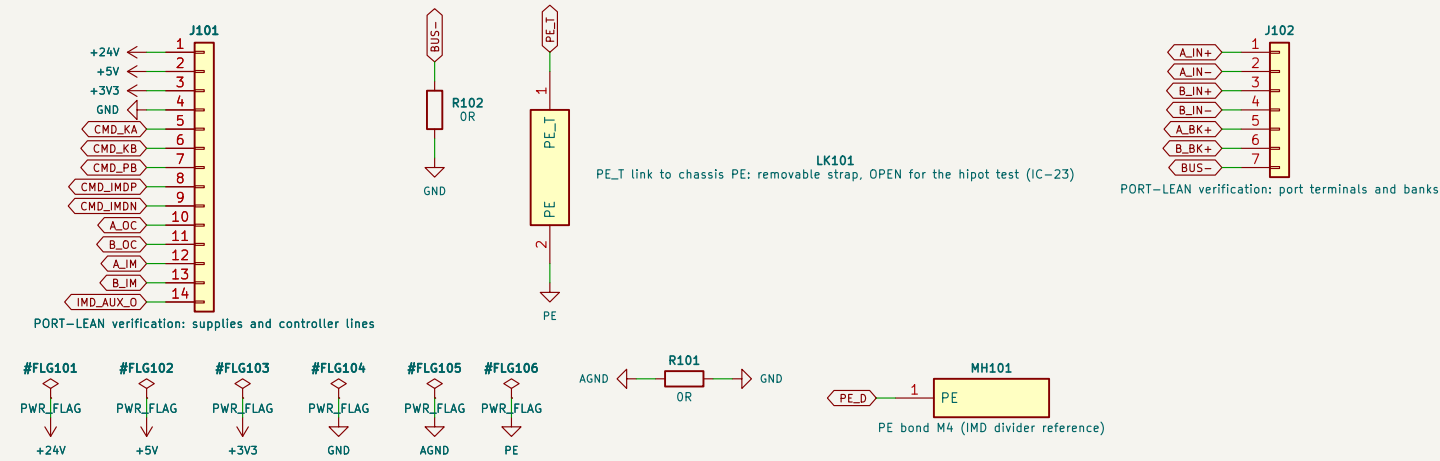
File: sheets/08_lean_imd.kicad_sch

Verification build of the reusable lean-port functions of gen/port.py (lean_port, varistor_y, lean_contactor, lean_precharge, lean_bleeder, lean_dividers, lean_shunt, lean_interlock, lean_imd). The product sheets live on the power board.

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 1
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: / File: PORT-LEAN.kicad_sch		
Title: Lean port sheets – Cover / Hierarchy		
Size: A3	Date: 2026-10-04	Rev: F0
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SUPPLIES AND COMMANDS



On the power board these come from the AUX-T1 live winding and the controller.

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Sheet 2
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /01 Supplies and commands/
File: 01_lean_io.kicad_sch

Title: Lean port sheets – Supplies and commands

Size: A3	Date: 2026-10-04	Rev: F0
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PORT A (PV SOURCE), D-044 LEAN PORT

Terminal slide A_{LN} / A_{JN} - bank A_{BK} - No fuse, no precharger; array loc. <= 1.25 x rating.
 Contactor opens <= 0.97 s, held above (port spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side AJN+ / AJN-, bank A8K+. No fuse, no precharge; array Isc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

—BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)

TV25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE, U_peff 3793 V pole-P and pole-pole at 1n 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA. Branch fuses ETI CH4x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNV65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 4

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /03 Port A sensing and interlocks/

File: 03_leanA_ctrl.kicad_sch

Title: Lean port sheets – Port A sensing and interlocks

Size: A3

Date: 2026-10-04

Rev: F0

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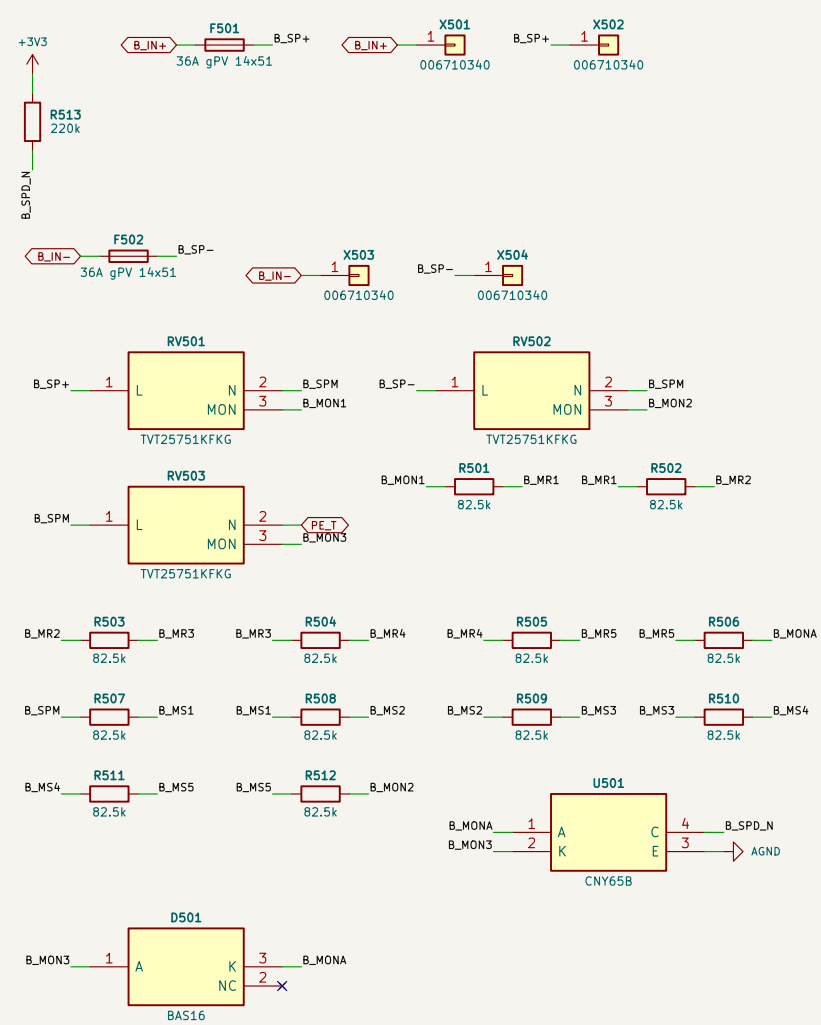
Id: 4/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'		
Sheet 5		
Schematics authored by Chinmoy Bhuyan		
Live side referenced to BUS-; basic HV-PE; functional inside the live side		
Sheet: /04 Port A coil drives/ File: 04_leanA_drive.kicad_sch		
Title: Lean port sheets – Port A coil drives		
Size: A3	Date: 2026-10-04	Rev: F0
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Terminal side B_{UN}+ / B_{UN}-, bank B_{BK}-. HPE501 250 A aR in both poles (breaks 1.25–50 kA only).
 Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

Terminal side B_{JN+} / B_{JN-}, bank B_{BK+}, HPIE501 250 A aR in both poles (breaks 1.25–50 kA only).
Circuit breaker opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.



TV125751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE_T. Up, eff. 3793 V pole-PE and pole-pole at in 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA, Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CRY65B

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 7
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /06 Port B sensing and interlocks/
File: 06_leanB_ctrl.kicad_sch

Title: Lean port sheets – Port B sensing and interlocks

Size: A3

Date: 2026-10-04

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Id: 7/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'		
Sheet 8		
Schematics authored by Chinmoy Bhuyan		
Live side referenced to BUS-; basic HV-PE; functional inside the live side		
Sheet: /07 Port B coil drives/ File: 07_leanB_drive.kicad_sch		
Title: Lean port sheets – Port B coil drives		
Size: A3	Date: 2026-10-04	Rev: F0
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	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 9

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /08 Insulation monitor/

File: 08_lean_imd.kicad_sch

Title: Lean port sheets – Insulation monitor

Size: A3

Date: 2026-10-04

Rev: F0

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